Catalog-wide curve-cover transfer assessment

All 2,091 current maps and all 664 EC1 bindings

2026-10-13

Outcome

Every current cover map and every linked EC1 representation now has an explicit v2 assessment. Classification is complete; implementation support is not. In particular, the complete inventory must not be read as a complete Jacobian-transfer pipeline.

The main results are:

- all 2,091 explicit cover formulas replay over their declared fields, subject to the field-validity assumptions stated below;
- all 160 binary maps have the implemented one-infinity genus-three Jacobian arithmetic and bounded rational-point-class transfer;
- all 134 binary EC1 bindings pass typed native tests for generator membership, `[r]G=O`, pullback validity, `[r]phi^*(G)=O`, composition, and degree-inverse recovery;
- those generator-level tests do **not** prove pullback additivity, all-element composition, injectivity, or image-order equality on the full generated subgroup; only the retained order-29 overlay has the stronger narrow exact-order and image-order certificate;
- all 1,931 odd-characteristic covers have verified genus-two geometry, but their even sextics have two rational infinity points and are outside the current one-infinity odd-characteristic Jacobian implementation;
- the binary arbitrary-class Jacobian norm is explicitly unsupported, and the odd-characteristic induced maps remain unimplemented and unknown; and
- no complete Jacobian relation solver, measured transfer advantage, or key recovery result is part of this assessment.

The canonical machine record is `cover-transfer-assessments.json`, generated by `build_cover_transfer_assessments.mjs` and validated against `cover-transfer-assessment.schema.json`. It uses only `supported`, `refuted`, `unknown`, and `not_applicable`. `Refuted` always refers to a concrete implementation or evidence claim; it never turns a missing implementation or failed bounded search into a theorem of nonexistence.

Original requirements and evidence status

Requested item	Status	Evidence boundary
Primary theorem and exact hypotheses	verified	Xarles v1, Corollary 3; same base field, characteristic not 2 or 3, higher genus at most 5, two independent maps
Audit crypto PR #1389 and cryptanalysis PR #290	verified historical audit	pinned heads and matching `quartic-ic` source hashes below; this is a non-hyperelliptic norm-projection experiment

Requested item	Status	Evidence boundary
Separate supported input models	implemented with explicit boundaries	binary and odd-characteristic constructors are separate; unsupported infinity configurations return a structured result
Constructive covers	supported for the two fixed catalog families	all 2,091 formulas are explicit and replayed; this is not a theorem-complete Xarles constructor
Genuine Jacobian arithmetic	partial	binary one-infinity sources are supported; all 1,931 odd two-infinity catalog sources remain outside the current backend
Pullback, pushforward, and subgroup transfer	partial	bounded binary point-class domains and 134 generator checks only; arbitrary binary norm and all odd induced maps remain open
EC1/ICV1 catalog evidence	implemented	exact memberships, typed object bindings, hashes, statuses, and four narrow overlays; classification is not support
Standalone executable	source verified; packaged scope incomplete	current-source tests cover 2,091 inputs; the published artifact was replayed on its bundled 592-row registry, not on all current inputs
Validation and shipping	partial	deterministic checks and native tests exist; independent divisor-class replay, local Lean kernel receipt, fresh package, and final CI gates remain open
Full Jacobian index-calculus solver	separate and not delivered	no genuine Jacobian relation collection, target decomposition, matched cost record, or end-to-end result

Population and profile coverage

The v2 ledger assigns every exact map UID to one map profile and every exact `[EC1, curve UID, map UID]` triple to one subgroup profile. Sorted memberships and domain-separated membership hashes make the coverage checkable rather than count-only.

Map profile	Maps	Source / map	What is established
Binary cubic, same field	160	genus 3, degree 3	geometry, one-infinity arithmetic, bounded typed transfer
Direct odd prime field	577	genus 2, degree 2	cover geometry only
Normalized odd prime field	30	genus 2, degree 2 plus degree-one target normalization	cover and normalization geometry only
Odd extension field	1,324	genus 2, degree 2 over the declared `GF(p^k)`	cover geometry over that field only
Total	**2,091**	**2,089 unique HC1 cover models**	**every map classified**

The two repeated cover objects are intentional shared models, not duplicate assessment rows. The 1,931 odd-

characteristic rows comprise 607 prime-field rows and 1,324 extension-field rows.

Subgroup profile	EC1 bindings	Established catalog-wide facts	Still open
Binary degree 3	134	exact identity joins, `gcd(3,r)=1`, typed generator-level transfer checks	additivity and all-element composition; injectivity and image order; proof-grade numeric exact order except order 29; arbitrary-result norm/recovery
Odd degree 2	530	exact identity joins and `gcd(2,r)=1`	generator/order certificates and every induced-map claim
Total	**664**	**every binding classified**	**unknowns remain explicit**

These are 664 EC1 representations on 659 curve models; five models have two distinct representations. The semantic verifier computes the effective `supported`, `unknown`, `refuted`, and `not_applicable` assignment totals from the generated ledger. Those are claim assignments, not counts of distinct maps, and no count of classified claims implies that the claims are supported.

Typed identity bindings are contexts, not capability claims

The ledger's explicit `typed_bindings` materialize 311 content-addressed declared-field records, 2,089 Jacobian contexts, and 2,091 typed curve-map bindings. A map binding joins the exact target ICV1 slug and model hash to its source-cover UID, source-Jacobian UID, and declared-field relationship. A Jacobian context records `Picard_degree_zero`, its dimension, its cover and field UIDs, and whether the source has one or two rational infinity points.

These records prevent accidental mixing of fields, source curves, infinity models, or maps. Their word `identity` is deliberately narrow: `same_declared_field_identity` says that the source-to-target field relationship is the identity, of relative degree one. The declared field itself may still be `GF(p^k)` with absolute extension degree greater than one, which the field binding retains. The identity does **not** assert that Jacobian arithmetic exists, that an induced pullback or pushforward is implemented, or that either is a homomorphism. Likewise, merely naming a Jacobian context does not promote its arithmetic. Capability status comes only from the applicable profile claim, result template, and any exact overlay.

Construction and geometry

Binary characteristic

For an ordinary binary target over its exact polynomial-basis field,

$$E : y^2 + xy = x^3 + ax^2 + b, \qquad b \neq 0,$$

put $d = \sqrt{b}$ and construct

$$H : v^2 + u^2 v = u^7 + au^4 + du, \qquad \phi(u, v) = (u^3, uv + d).$$

The native checker verifies $d^2 = b$, the defining-equation identity, affine smoothness, the infinity chart, genus three, degree three, and separability. The binary field polynomial is checked by the exact Rabin criterion. The details and proof are in `COVERS.md` and the implementation is in `checker.rs`.

Odd characteristic

After any recorded degree-one Montgomery or Edwards normalization, write the short target as

$$E : y^2 = F(x) = x^3 + ax + b.$$

Choose the first $c \in \{0, 1, 2, 3\}$ with $F(c) \neq 0$, and construct

$$H : v^2 = F(u^2 + c), \qquad \phi(u, v) = (u^2 + c, v).$$

The checker verifies the equation identity, the monic squarefree sextic, genus two, two rational points at infinity, degree two, separability, and the recorded target-model identities. This construction works over the same declared prime field or polynomial-basis extension field; it is not a descent to a proper subfield.

For odd characteristic, `p` passes a fixed 20-base probable-prime screen but is retained as a registry input assumption, not a primality proof. Extension moduli pass the exact Rabin irreducibility criterion *conditional on that prime-field assumption*. Thus "verified over the declared field" has this explicit boundary. The catalog checker does not certify group orders, generators, or EC1 subgroup parameters.

The independent semantic replay in `verify_assessment.mjs` checks every content join, profile predicate, membership hash, and all 2,091 finite-field map formulas. It does not independently prove odd `p` primality, extension-modulus irreducibility without that assumption, EC1 subgroup orders, or computational advantage.

Original theorem audit

The broader source is Xavier Xarles, *Hyperelliptic curves covering an elliptic curve twice*, arXiv:1303.4220v1, Corollary 3. Its statement applies to an elliptic curve over an arbitrary field K of characteristic different from 2 and 3. It gives a hyperelliptic curve H/K of genus at most five with two independent K-defined maps to the elliptic curve; it does not require first extending the base field.

For $j \neq 0, 1728$, the paper displays a genus-five construction with two degree-three maps. Degree three together with characteristic different from three makes those displayed generic maps separable. Separability is not a separately stated hypothesis or conclusion of Corollary 3, however, and the exceptional `j` cases invoke earlier genus-two results whose maps were not separately audited here for separability. The corollary is therefore an existence theorem with explicit generic formulas, not a single uniform executable constructor and certificate for every input model. Its exceptional-case citations must not be silently upgraded into the repository's implementation evidence.

"Hyperelliptic" here means the higher-genus setting, genus at least two. The tautological genus-one reading in which an elliptic curve covers itself is not the content being used. The repository's two fixed families above are separate constructive results: they give one explicit map per supported catalog model and do not claim Xarles's pair of independent maps.

Existing `quartic-ic` experiment audit

Crypto PR #1389 and cryptanalysis PR #290 were audited at these immutable revisions:

- crypto PR #1389 head: `6c6247bde5e585642773ed5b6b933b4ec4c6fd4f`
- crypto PR #1389 merge: `7ea1db7a6b4b6962f3fa38d70ac65300d25ceb4b`
- cryptanalysis PR #290 head: `3b14ec30de6c79fbc9dd72f4ed3eb8ea19d5a642`
- cryptanalysis PR #290 merge: `54cbf1dc27879b9bd441df7b64e2c629e9d9d9ea`

At both heads, the quartic kernel, CLI, and retained receipt were byte-identical with these SHA-256 values:

```
kernel   54ef65c7325a7c850741a8def84ee2b2ceddcc3689a0879afd2796eb7cb01e52
CLI      0f71936014b3ff54dacc9dc6b642c88bea60b70a658f6e42c467aab73f11a069
receipt  d3de82409e6fe82a59e74425916ed820ac5f8912e74046f41468cef56c01a967
```

The immutable October 6 audit records the retrieval and replay details.

The experiment uses the smooth plane quartic

$$C : v^4 = x^3 + ax + b, \qquad \pi(x, v) = (x, v^2),$$

over bounded small prime fields. A smooth plane quartic has genus three and is **non-hyperelliptic**. `quartic-ic` certifies complete line sections, then forms its linear system after elliptic norm projection. Its fixture's passing tests support that bounded projected experiment; they do not construct a hyperelliptic cover, implement divisor-class arithmetic in the full quartic Jacobian, or establish a Jacobian index-calculus solver. Deck-conjugate divisors may share an elliptic norm while retaining different Prym components, so elliptic projected relations cannot be relabeled as full-Jacobian relations.

Jacobians and transfer maps

Binary supported domain

`hyperelliptic.rs` implements checked canonical reduced Mumford divisors for the binary one-infinity source family: validity, reduction, equality, negation, addition, and scalar multiplication. Its tests include identity, exceptional divisors, shared support, inverses, and exhaustive small-field fixtures.

`cover_transfer.rs` implements:

- pullback of rational elliptic point classes `[P]-[O]`, including infinity and the exceptional ramified fiber;
- pushforward of rational source-point divisors and the implementation's own typed pullback witnesses; and
- `phi_* phi^* = [3]` on that typed point-class domain.

It deliberately does **not** implement a norm on an arbitrary reduced class of `Jac(H)`. The public negative-control path returns `Unsupported(ArbitraryJacobianPushforward)`. That refutes a claim about the current API, not the mathematical existence of the norm. This is tracked in `binary-cover-arbitrary-jacobian-norm-missing.md`.

The 134 binary subgroup checks

For every exact linked binary EC1 triple, the native batch test decodes the catalog generator and verifies:

1. the generator lies on the exact target and is nonidentity;
2. `[r]G=O` for the catalog integer `r`;
3. `phi^*(G)` is a valid nonidentity reduced divisor;
4. `[r]phi^*(G)=O`;
5. the typed composition gives `phi_*phi^*(G)=[3]G`; and
6. multiplication by the inverse of 3 modulo `r` recovers `G` on this typed route.

Because `[r]G=O`, the actual order of `G` divides `r`, and the checked modular inverse calculation recovers this one tested generator from its typed composition result. That is not enough to prove a homomorphism or injectivity on all of `<G>`: the batch test does not establish pullback additivity or the composition identity for every subgroup element through an independent map oracle. Catalog-wide kernel intersection and actual image/source-order equality therefore remain `unknown` and conditional. It also does **not** prove `ord(G)=r`. Proof-grade numeric equality with the catalog order remains `unknown` for 133 of the 134 binary bindings; the retained $\mathbf{F}_{2^7}$ order-29 overlay is the one narrow exception.

The elementary group-theoretic core is formalized in `CoverTransfer.lean`: if a finite subgroup has exact cardinality `r`, `push(pull(x))=d*x` on that subgroup, and `Nat.Coprime d r`, then the restricted pullback is injective. The written proof is immediate: equal pullbacks give equal `d`-multiples after applying pushforward, while multiplication by `d` is bijective on a finite group whose cardinality is coprime to `d`.

That Lean theorem assumes exact cardinality and the composition identity; it constructs neither map and does not promote the 134 catalog rows by itself. The additional bridge from `[r]G=O` to `ord(G) | r`, and hence coprimality with the actual generated-subgroup order, is used in the native certificate but is not a separately formalized Lean corollary here; additivity and all-element composition are still missing. The sources pin Lean 4.34.1 and Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Local kernel replay remains blocked: `lake exe cache`

`get` stopped at 2,536 of 8,908 files when the sandbox ran out of space, so `lake build` and the axiom audit have not been executed locally. Until CI completes those commands, the source is a formalization candidate, not a kernel-verified proof receipt. The exact boundary is recorded in the Lean README.

Odd-characteristic boundary

Every odd-characteristic source is a degree-six genus-two curve with two rational infinity points. The current `prime_hyperelliptic` backend requires a monic polynomial of degree `2g+1` and one rational infinity. Consequently:

- the claim that the current backend supports these 1,931 sources is `refuted`;
- balanced/two-infinity divisor arithmetic, pullback, pushforward, and `phi_*phi^*=[2]` remain `unknown`; and
- for the 530 linked odd EC1 records, the exact identity join and numerical `gcd(2,r)=1` are supported, but generator, order, kernel, image-order, and recovery certificates remain unknown.

This implementation gap does not refute the mathematical Jacobian or its induced Picard-group maps. It is tracked in `prime-cover-balanced-jacobian-and-induced-maps-missing.md`. The remaining proof-grade subgroup obligations are tracked separately in `catalog-cover-subgroup-certificates-incomplete.md`.

Exact overlays retained from the prior audit

The v2 ledger keeps four narrow overlays rather than extrapolating them:

Overlay	Added evidence	Boundary
`icv1-f2m7-t13-616700dd` degree-three map	detailed geometry replay	does not widen the typed transfer domain
`icv1-f2m7-t13-616700dd` order-29 subgroup	exact source and image order	only that one EC1/map triple
`icv1-fp10-t5-192cb216` degree-two map	detailed geometry replay	no induced odd Jacobian map
`icv1-fp10-t5-192cb216` order-823 identity	exact catalog identity retained	no preservation claim follows

The prior October 6 evidence remains immutable in `evidence.json`. The v2 generator reuses it only for these exact overlays. The stale `jacobian-transfers.json` formula screen is not imported as a map, order, or recovery certificate.

Source, packaged binary, and browser boundary

The current source allowlist now accepts all four catalog input profiles. The catalog-wide constructor test reaches the standalone `construct` path for all 2,091 current models. This is source-level evidence at the current revision, not a published-artifact claim.

The independently replayed published binary `v0.1.0-main.1371` predates that allowlist repair. Direct replay establishes that it runs against its own bundled 592-row registry. Separately, static comparison of its pinned source allowlist with the current model partition places 737 forms inside its syntactic domain and 1,354 outside: the latter are 1,324 extension-field rows and 30 normalized Montgomery/Edwards rows. No exact batch receipt ran all 2,091 current inputs against that archived executable. The artifact is therefore **not** evidence of a catalog-wide current binary. Exact asset hashes, directly executed commands, and the static-classification boundary are preserved in `release_replay.txt`.

The generated curve browser remains deliberately unchanged. Its 2,091 cards report geometry as `verified_over_declared_field`, retain `subgroup_transfer: not_tested`, and keep DLP advantage null. The v2 sidecar does not silently turn those geometry cards into transfer badges.

The current offline cryptanalysis parity rerun reported `hyperelliptic mirror manifest and crypto source: verified` for exactly seven manifest records. This establishes byte-level parity only for the listed core files. It

excludes the standalone CLI, Cargo configuration, catalogs, schema, assessment ledger, workflows, and package artifacts, and it does not check the manifest's recorded commit against the checkout's actual `HEAD`. It therefore does not establish that the present standalone CLI allowlist repair has been packaged or mirrored at a new exact tip.

Validation controls

Control	Status	Scope and limit
Exact identity joins	`supported`	all 2,091 map UIDs and 664 EC1/curve/map triples; identity alone claims no arithmetic
Cover geometry	`supported`	all 2,091 fixed-family rows; equations, smoothness, genus, degree, separability, and infinity behavior
Internal Jacobian arithmetic	`supported` on stated models	binary and separate odd-prime one-infinity fixtures; not the 1,931 odd two-infinity cover sources
Independent arithmetic/map oracle	`unknown`	no separate CAS or independently implemented divisor-class replay across the catalog
Binary catalog generators	`supported` on tested generators	all 134 triples pass the listed membership, annihilation, pullback, composition, and inverse-degree checks; no subgroup-wide promotion
Odd cover/backend compatibility	`refuted` as a current implementation claim	the one-infinity backend cannot accept the 1,931 even-sextic sources; mathematical induced maps remain `unknown`
Published artifact on all current inputs	`refuted` as evidence claim	only its bundled 592 rows were directly replayed; 737/1,354 is a static allowlist classification, not a 2,091-command receipt
Cross-repository core parity	`supported` on manifest-listed scope	offline rerun verified exactly seven core-file records; CLI, Cargo, catalogs, schema, ledger, workflows, packages, and actual-HEAD binding are excluded
Coordinate commutation and proper-subfield descent	`unknown`	normalization geometry is checked, but induced-divisor commutation and descent are not
Paired performance	`not_applicable`	no performance or speedup experiment was conducted

Cost and solver boundary

No performance experiment was conducted. Every measured-value field below is therefore null; correctness-test elapsed time is not an algorithmic cost.

Phase	Status	Value	Why
Construction and failed searches	`unknown`	null	fixed constructors have no calibrated search ledger

Phase	Status	Value	Why
Duplicate candidates	`unknown`	null	no candidate-deduplication accounting
Geometry verification	`unknown`	null	replayable but uncalibrated
Jacobian arithmetic	`unknown`	null	correctness tests only
Pullback, pushforward, recovery	`unknown`	null	bounded binary path; no complete measured conversion
Reusable preprocessing	`unknown`	null	no amortization boundary
Peak memory and total runtime	`unknown`	null	not measured end to end
Matched generic baseline	`unknown`	null	not run
End-to-end speedup	`unknown`	null	numerator and denominator absent
Relations, linear algebra, target decomposition	`not_applicable`	null	separate solver not implemented or run

The full Jacobian index-calculus solver remains a separate deliverable. Its acceptance gate must require genuine Jacobian relations, verified linear algebra and target decomposition, independently checked outputs, explicit parameter limits, and complete accounting for preprocessing, failed searches, duplicates, memory, and total runtime. Elliptic norm projection is not that solver.

Visual summary

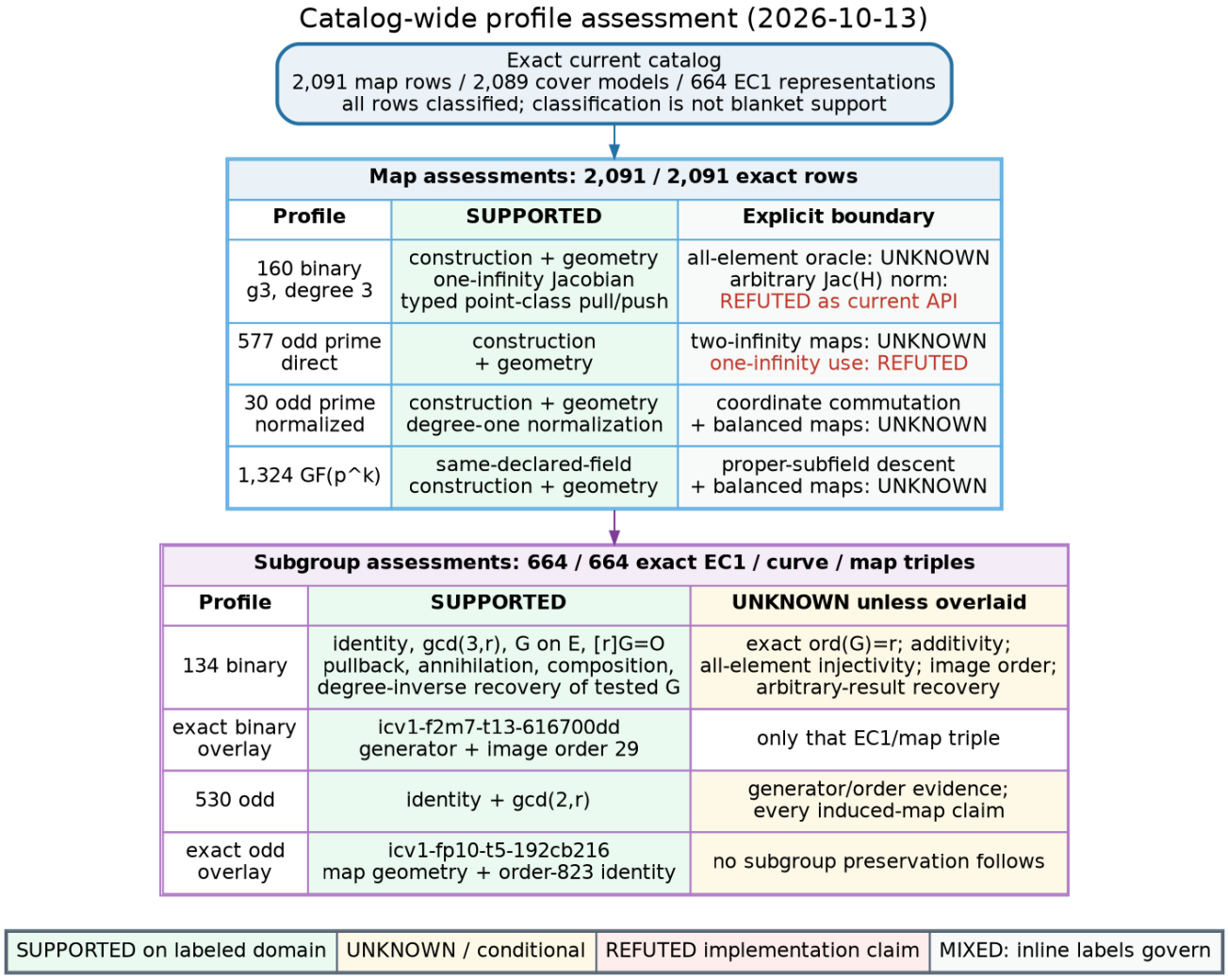


Figure 1: Catalog-wide map and subgroup classification. Green denotes supported claims on the labeled domain, orange unknown obligations, red refuted current implementation claims, and neutral cells contain explicitly mixed statuses governed by their inline labels.

Editable source: `typed_correspondence.dot`. Vector rendering: `typed_correspondence.svg`.

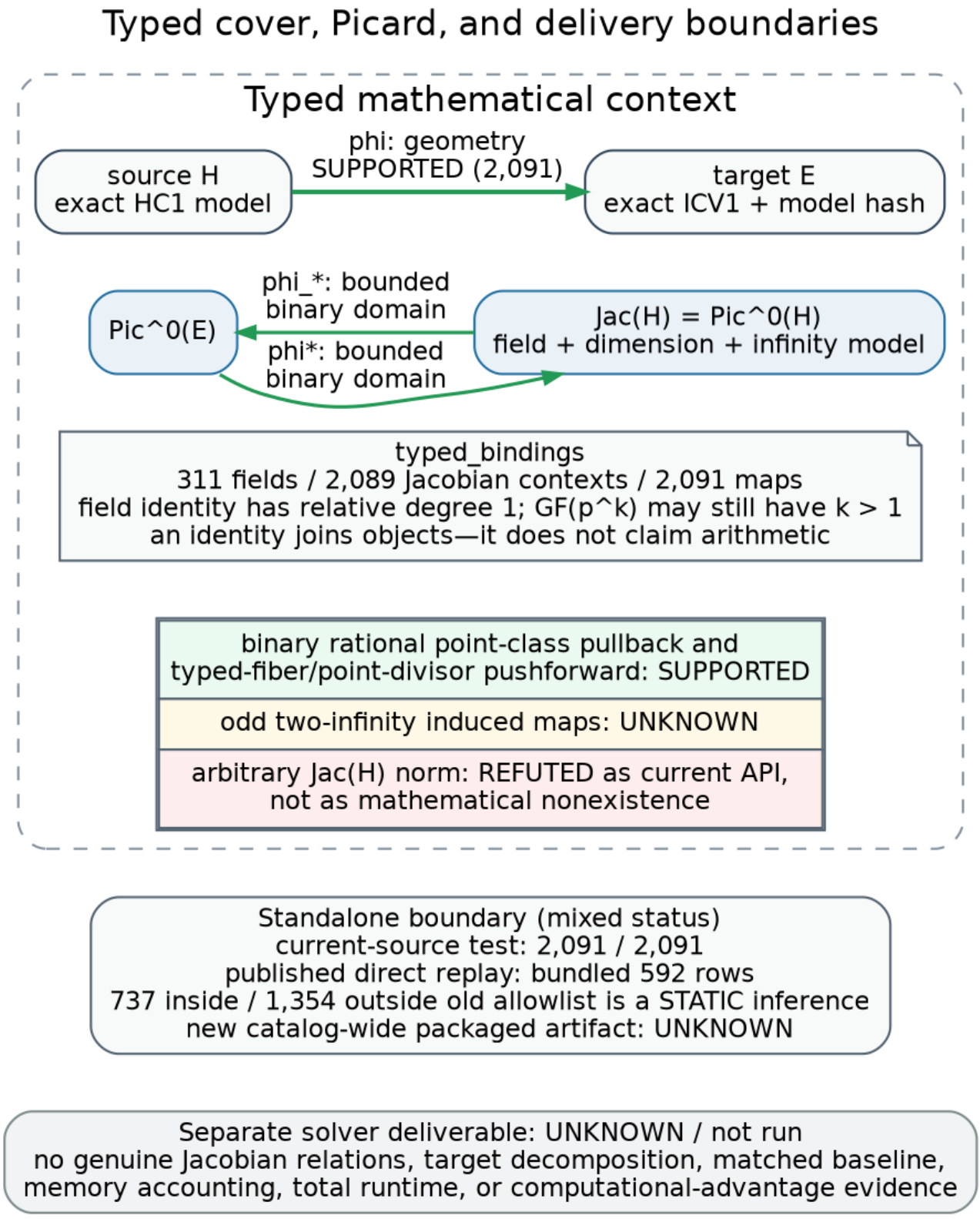


Figure 2: Typed curve/Picard context and delivery boundaries. Green arrows exist only on the bounded domains printed beside them. An identity binding does not claim arithmetic, a transfer implementation, or a solver result.

Editable source: `typed_transfer_context.dot`. Vector rendering: `typed_transfer_context.svg`. No edge in either figure denotes an unlisted solver result.

Reproduction

Generate and verify the exact catalog-wide ledger:

```
node scripts/build_cover_transfer_assessments.mjs --check
python -m jsonschema \
  -i docs/curves/cover-transfer-assessments.json \
  docs/curves/cover-transfer-assessment.schema.json
node research/cover_transfer_assessment_20261013/verify_assessment.mjs
```

Replay the native geometry, arithmetic, and subgroup checks:

```
cargo test --release --locked --bin curve_cover_check
cargo test --release --locked --lib hyperelliptic
cargo test --release --locked --lib binary_ecc::cover_transfer::tests
cargo test --release --locked --bin hyperelliptic-cover -- \
  --test-threads=1
cargo test --release --locked --bin hyperelliptic-cover \
  every_current_catalog_model_reaches_the_standalone_constructor
cargo test --release --locked --bin hyperelliptic-cover \
  every_binary_ec1_generator_has_checked_typed_transfer_hypotheses
```

The complete standalone-binary suite was run in the pinned Rust 1.90.0 container and reported 11 passed, 0 failed. Rebuild and audit the Lean package when sufficient cache space is available:

```
cd research/cover_transfer_assessment_20261013/lean
lake exe cache get
lake build
lake env lean AxiomAudit.lean
```

Regenerate the visual and PDF after any reported count, status, or conclusion changes:

```
cd research/cover_transfer_assessment_20261013
dot -Tsvg typed_correspondence.dot -o typed_correspondence.svg
dot -Tpng -Gdpi=150 typed_correspondence.dot -o typed_correspondence.png
dot -Tsvg typed_transfer_context.dot -o typed_transfer_context.svg
dot -Tpng -Gdpi=150 typed_transfer_context.dot \
  -o typed_transfer_context.png
pandoc README.md \
  --from=markdown+tex_math_single_backslash \
  --pdf-engine=pdflatex \
  -o cover_transfer_assessment_20261013.pdf
pdfinfo cover_transfer_assessment_20261013.pdf
```

Open obligations

These obligations feed WS1 and WS2 of the proposed comprehensive elliptic-curve private-key recovery audit. That program link does not change any status below and does not turn the cover assessment into a solver or advantage result.

1. Implement and independently verify a binary arbitrary-class norm.
2. Implement balanced/two-infinity odd-characteristic Jacobian arithmetic and induced maps over both prime and declared extension fields.
3. Add proof-grade exact-order evidence and map-bound certificates wherever a numeric EC1 order or image-order claim is desired.
4. Replay map additivity and composition through an independent divisor-class implementation.
5. Add coordinate-isomorphism commutation and proper-subfield descent controls.
6. Publish and replay a standalone artifact that accepts all 2,091 current model forms.
7. Refresh the legacy formula inventory without promoting modeled routes to certificates.
8. Keep the full solver and every cost claim behind its separate acceptance gates.

Exploration boundary

This work assessed supplied public catalog constructions. It did not search production targets, recover a private key, run a discrete-logarithm campaign, prove minimal genus or degree, exhaust all cover families, or infer security from an unsupported path. A failed construction outside the supported families remains a structured unsupported result, never proof that no cover exists.